

Matrix transformations in 2D

Key Points

For a matrix, M , to visualise transforms not on the standard list, use the transform on unit vectors

$$\begin{pmatrix} 1 \\ 0 \end{pmatrix}, \begin{pmatrix} 0 \\ 1 \end{pmatrix}$$

Some transforms can be found in the formula booklet.

A shear transforms stretches a coordinate dependent on the other coordinate.

Basic Skills

Q1. Compute the following matrix calculations

a) $\begin{pmatrix} 2 & 3 \\ 3 & 6 \end{pmatrix} \begin{pmatrix} 3 \\ 1 \end{pmatrix}$ b) $\begin{pmatrix} -1 & 4 \\ 3 & 0 \end{pmatrix} \begin{pmatrix} 0 \\ -1 \end{pmatrix}$ c) $\begin{pmatrix} 5 & 0 & -5 \\ 6 & 7 & 3 \\ -4 & 3 & 0 \end{pmatrix} \begin{pmatrix} 1 \\ -3 \\ 4 \end{pmatrix}$

Q2. Write a matrix for the following transforms

- a) A stretch in x by a scale factor of 3
- b) A rotation anticlockwise by 90°
- c) A rotation clockwise by 270°
- d) An enlargement of scale factor $\frac{1}{2}$
- e) An reflection in the line $y = x$

Competency Skills

Q3. Describe the following matrix transformations

a) $\begin{pmatrix} \cos 10^\circ & -\sin 10^\circ \\ \sin 10^\circ & \cos 10^\circ \end{pmatrix}$ b) $\begin{pmatrix} 4 & 0 \\ 0 & 4 \end{pmatrix}$ c) $\begin{pmatrix} 1 & 0 \\ 0 & 7 \end{pmatrix}$ d) $\begin{pmatrix} 3 & 0 \\ 0 & 1 \end{pmatrix}$ e) $\begin{pmatrix} 1 & 6 \\ 0 & 1 \end{pmatrix}$ f) $\begin{pmatrix} 1 & 0 \\ 2 & 1 \end{pmatrix}$ g) $\begin{pmatrix} \frac{\sqrt{3}}{2} & -\frac{1}{2} \\ \frac{1}{2} & \frac{\sqrt{3}}{2} \end{pmatrix}$

Q4. Write the matrix transformation for a reflection in the line $y = -2x$

Q5. Is a translation possible to write as a matrix multiplication transform

Q6. A quadrilateral ABCD where $A=(0,2), B=(0,5), C=(3,5), D=(1,2)$ is sheared from the y -axis making the rectangle $AB'C'D'$ where $C'=(3,8)$ and $D'=(1,3)$ find the matrix representation of the shear

Q7. Determine the matrix which shears from the x -axis and maps $(0,1)$ to $(k,1)$ for any constant k

Q8. State a matrix transformation which maps all points in the 2-D plane to a single point

Advanced Skills

Q9. Find a matrix transformation which maps all points in 2D space onto the line $y = x$

Q10. The transform, M , of 30° anti-clockwise is followed by N , a stretch parallel to the x -axis of scale factor 4 find the matrix which represents the final transformation by remultiplying M by N

Q11. Find where the line of points $y = 2x$ is transformed to under the transform, $\begin{pmatrix} 4 & 8 \\ 1 & 1 \end{pmatrix}$

Q12. Show that the area scale factor is 2 for the transform $\begin{pmatrix} 2 & 2 \\ 1 & 2 \end{pmatrix}$ by using the unit square

Q13. Find the transformation which reverses the transform of $\begin{pmatrix} 3 & -4 \\ 1 & 2 \end{pmatrix}$

Extension

Q14. Decide what makes a transformation irreversible and state its conditions

Hint: If you know about matrix inverses consider when they don't exist